# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** $x^2 + 3x + 2 = 0$ | Degree must be 2 |
| 2 | **(d)** 2 | At $x = 2$ the left side equals $0$, so 2 is a root |
| 3 | **(b)** -3 | Compare with $ax^2 + bx + c = 0$ |
| 4 | **(a)** 2 and 3 | Put each factor equal to zero |
| 5 | **(c)** 1 | $D = b^2 - 4ac$ |
| 6 | **(a)** two distinct real roots | $D = b^2 - 4ac = 1$ |
| 7 | **(a)** $x^2 - 5x + 6 = 0$ | Degree must be 2 |
| 8 | **(d)** 3 | At $x = 3$ the left side equals $0$, so 3 is a root |
| 9 | **(d)** 7 | Compare with $ax^2 + bx + c = 0$ |
| 10 | **(a)** -1 and 4 | Put each factor equal to zero |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Put $x = 2$: LHS $= 0$ (1); equals RHS 0, so it is a root (1). | 1+1 |
| 12 | $x - 4 = 0$ or $x - 5 = 0$ (1); $x = 4$ or $x = 5$ (1). | 1+1 |
| 13 | $D = (-6)^2 - 4(1)(8) = 4$ (1); two distinct real roots (1). | 1+1 |
| 14 | Put $x = 3$: LHS $= -2$ (1); not equal to 0, so it is not a root (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Split the middle term: $-5 = -3 - 2$ because $ac = 6$ (1); regroup and factorise (1); roots $x = 2, 3$ (1). | 1+1+1 |
| 16 | $a = 2$, $b = -4$, $c = 3$ (1); $D = -8$ (1); no real roots (1). | 1+1+1 |
| 17 | Let integers be $x$ and $x + 1$: $x(x + 1) = 20$ (1); $x^2 + x - 20 = 0$ (1); $(x - 4)(x + 5) = 0$, $x = 4$ (1); integers 4 and 5 (1). | 1+1+1+1 |
| | **OR:** Split the middle term: $-4 = -5 + 1$ because $ac = -5$ (1); regroup and factorise (1); roots $x = -1, 5$ (1). | 1+1+1 |
| 18 | $a = 1$, $b = -8$, $c = 16$ (1); $D = 0$ (1); two equal real roots (1). | 1+1+1 |
| | **OR:** Let integers be $x$ and $x + 1$: $x(x + 1) = 42$ (1); $x^2 + x - 42 = 0$ (1); $(x - 6)(x + 7) = 0$, $x = 6$ (1); integers 6 and 7 (1). | 1+1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Split the middle term: $-7 = -6 - 1$ because $ac = 6$ (1); regroup and factorise (1); roots $x = \frac{1}{2}, 3$ (1). | 1+1+1 |
| | **OR:** Breadth $x$, length $x + 5$: $x(x + 5) = 66$ (1); $x^2 + 5x - 66 = 0$ (1); $(x - 6)(x + 11) = 0$ (1); $x = 6$ m (breadth), length 11 m (reject negative) (2). | 1+1+1+2 |
| 20 | Split the middle term: $-1 = -3 + 2$ because $ac = -6$ (1); regroup and factorise (1); roots $x = -\frac{2}{3}, 1$ (1). | 1+1+1 |
| | **OR:** Breadth $x$, length $x + 4$: $x(x + 4) = 96$ (1); $x^2 + 4x - 96 = 0$ (1); $(x - 8)(x + 12) = 0$ (1); $x = 8$ m (breadth), length 12 m (reject negative) (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Writing the formula as $\dfrac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ or dividing only part of the numerator by $2a$.
- Forgetting the sign of b: for $x^2 - 5x + 6 = 0$, $-b = +5$.
- Not shifting everything to one side to get the standard form $ax^2 + bx + c = 0$ before using $D$ or factorising.
- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x - 2) = 3x$).
- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D = 0$ means no roots; it means two *equal* real roots.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 1 | $D = b^2 - 4ac$ |
| 2 | **(d)** two equal real roots | $D = b^2 - 4ac = 0$ |
| 3 | **(b)** $x^2 + 4x + 1 = 0$ | Degree must be 2 |
| 4 | **(d)** 2 | At $x = 2$ the left side equals $0$, so $2$ is a root |
| 5 | **(b)** -4 | Compare with $ax^2 + bx + c = 0$ |
| 6 | **(a)** 3 and 5 | Put each factor equal to zero |
| 7 | **(b)** 0 | $D = b^2 - 4ac$ |
| 8 | **(a)** no real roots | $D = b^2 - 4ac = -11$ |
| 9 | **(c)** $x^2 + 7x - 2 = 0$ | Degree must be 2 |
| 10 | **(c)** -1 | At $x = -1$ the left side equals $0$, so $-1$ is a root |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $x + 3 = 0$ or $x + 4 = 0$ (1); $x = -3$ or $x = -4$ (1). | 1+1 |
| 12 | $D = (-4)^2 - 4(1)(4) = 0$ (1); two equal real roots (1). | 1+1 |
| 13 | Put $x = 1$: LHS $= 0$ (1); equals RHS 0, so it is a root (1). | 1+1 |
| 14 | $x - 2 = 0$ or $x + 5 = 0$ (1); $x = 2$ or $x = -5$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Split the middle term: $-10 = -6 - 4$ because $ac = 24$ (1); regroup and factorise (1); roots $x = 4, 6$ (1). | 1+1+1 |
| 16 | $a = 3$, $b = 5$, $c = -2$ (1); $D = 49$ (1); two distinct real roots (1). | 1+1+1 |
| 17 | Let integers be $x$ and $x + 1$: $x(x + 1) = 72$ (1); $x^2 + x - 72 = 0$ (1); $(x - 8)(x + 9) = 0$, $x = 8$ (1); integers 8 and 9 (1). | 1+1+1+1 |
| | **OR:** Split the middle term: $-4 = -7 + 3$ because $ac = -21$ (1); regroup and factorise (1); roots $x = -3, 7$ (1). | 1+1+1 |
| 18 | $a = 1$, $b = 6$, $c = 10$ (1); $D = -4$ (1); no real roots (1). | 1+1+1 |
| | **OR:** Let integers be $x$ and $x + 1$: $x(x + 1) = 132$ (1); $x^2 + x - 132 = 0$ (1); $(x - 11)(x + 12) = 0$, $x = 11$ (1); integers 11 and 12 (1). | 1+1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Split the middle term: $1 = -3 + 4$ because $ac = -12$ (1); regroup and factorise (1); roots $x = \frac{3}{2}, -2$ (1). | 1+1+1 |
| | **OR:** Breadth $x$, length $x + 3$: $x(x + 3) = 130$ (1); $x^2 + 3x - 130 = 0$ (1); $(x - 10)(x + 13) = 0$ (1); $x = 10$ m (breadth), length 13 m (reject negative) (2). | 1+1+1+2 |
| 20 | Split the middle term: $-7 = -6 - 1$ because $ac = 6$ (1); regroup and factorise (1); roots $x = \frac{1}{3}, 2$ (1). | 1+1+1 |
| | **OR:** Breadth $x$, length $x + 6$: $x(x + 6) = 91$ (1); $x^2 + 6x - 91 = 0$ (1); $(x - 7)(x + 13) = 0$ (1); $x = 7$ m (breadth), length 13 m (reject negative) (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Writing the formula as $\dfrac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ or dividing only part of the numerator by $2a$.
- Forgetting the sign of b: for $x^2 - 5x + 6 = 0$, $-b = +5$.
- Not shifting everything to one side to get the standard form $ax^2 + bx + c = 0$ before using $D$ or factorising.
- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x - 2) = 3x$).
- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D = 0$ means no roots; it means two *equal* real roots.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 6 | Compare with $ax^2+bx+c=0$ |
| 2 | **(d)** -2 and 6 | Put each factor equal to zero |
| 3 | **(b)** 16 | $D=b^2-4ac$ |
| 4 | **(b)** two equal real roots | $D=b^2-4ac=0$ |
| 5 | **(c)** $x^2+2x+9=0$ | Degree must be 2 |
| 6 | **(a)** 1 | At $x=1$ the left side equals $0$, so $1$ is a root |
| 7 | **(b)** -8 | Compare with $ax^2+bx+c=0$ |
| 8 | **(d)** 1 and 7 | Put each factor equal to zero |
| 9 | **(a)** -11 | $D=b^2-4ac$ |
| 10 | **(d)** $x^2-x+3=0$ | Degree must be 2 |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $D=(1)^2-4(2)(3)=-23$ (1); no real roots (1). | 1+1 |
| 12 | Put $x=-4$: LHS $=0$ (1); equals RHS 0, so it is a root (1). | 1+1 |
| 13 | $x-6=0$ or $x+1=0$ (1); $x=6$ or $x=-1$ (1). | 1+1 |
| 14 | $D=(-7)^2-4(3)(2)=25$ (1); two distinct real roots (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Split the middle term: $-9=-8-1$ because $ac=8$ (1); regroup and factorise (1); roots $x=1,8$ (1). | 1+1+1 |
| 16 | $a=5$, $b=-10$, $c=5$ (1); $D=0$ (1); two equal real roots (1). | 1+1+1 |
| 17 | Let integers be $x$ and $x+1$: $x(x+1)=182$ (1); $x^2+x-182=0$ (1); $(x-13)(x+14)=0$, $x=13$ (1); integers 13 and 14 (1). | 1+1+1+1 |
| | **OR:** Split the middle term: $6=2+4$ because $ac=8$ (1); regroup and factorise (1); roots $x=-4,-2$ (1). | 1+1+1 |
| 18 | $a=2$, $b=7$, $c=3$ (1); $D=25$ (1); two distinct real roots (1). | 1+1+1 |
| | **OR:** Let integers be $x$ and $x+1$: $x(x+1)=240$ (1); $x^2+x-240=0$ (1); $(x-15)(x+16)=0$, $x=15$ (1); integers 15 and 16 (1). | 1+1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Split the middle term: $-5=-8+3$ because $ac=-24$ (1); regroup and factorise (1); roots $x=-\frac{3}{4},2$ (1). | 1+1+1 |
| | **OR:** Breadth $x$, length $x+8$: $x(x+8)=153$ (1); $x^2+8x-153=0$ (1); $(x-9)(x+17)=0$ (1); $x=9$ m (breadth), length 17 m (reject negative) (2). | 1+1+1+2 |
| 20 | Split the middle term: $-3=-5+2$ because $ac=-10$ (1); regroup and factorise (1); roots $x=\frac{5}{2},-1$ (1). | 1+1+1 |
| | **OR:** Breadth $x$, length $x+5$: $x(x+5)=204$ (1); $x^2+5x-204=0$ (1); $(x-12)(x+17)=0$ (1); $x=12$ m (breadth), length 17 m (reject negative) (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Writing the formula as $\dfrac{-b\pm\sqrt{b^2+4ac}}{2a}$ or dividing only part of the numerator by $2a$.
- Forgetting the sign of b: for $x^2-5x+6=0$, $-b=+5$.
- Not shifting everything to one side to get the standard form $ax^2+bx+c=0$ before using $D$ or factorising.
- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x-2)=3x$).
- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D=0$ means no roots; it means two *equal* real roots.

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